ECE 329 **Homework 11** **Due: Mon, Dec 15, 2025, 11:59PM**

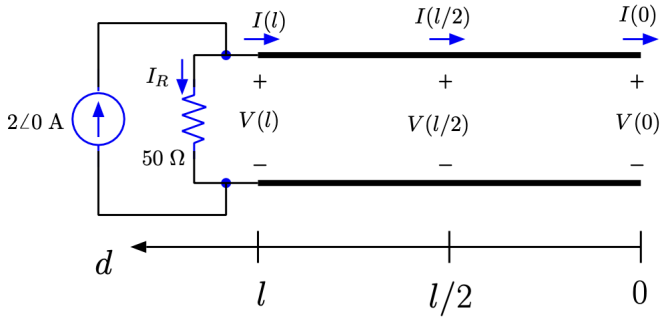
1. (10pts) Consider a lossless TL which is open circuited at both ends. If $l = 10$ m is the length of the line, and $v = \frac{1}{3}c = 10^8$ m/s for the line,

   a) (3pts) What are all the resonance frequencies of the line — frequencies at which source-free *oscillations* (standing waves) of voltage and current are sustained on the line — expressed in MHz units? **Hint:** to satisfy the "open" boundary conditions at both ends of the line, we want the current phasor to vanish at $z = 0$ and $z = l$ when the oscillation frequency is resonant.

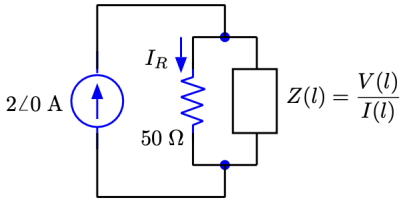
   b) (4pts) Sketch the shapes of current magnitude $|I(z)|$ vs z for the line corresponding to the *three lowest resonance frequencies*. Label each plot clearly and explain each briefly.

   c) (3pts) Repeat (b) for voltage magnitude $|V(z)|$ vs z.

2. (10pts) Consider a transmission line segment of propagation velocity $v = \frac{2}{3}c = 2 \times 10^8$ m/s, characteristic impedance $Z_o = 50\,\Omega$, and length l. As shown in figure (a) below, the segment is connected *in parallel* with a 50 Ω **resistor** and an ideal **current source** $i(t) = \mathrm{Re}\{Ie^{j\omega t}\}$, where $I = 2\angle 0$ A is the source current phasor and $\omega = \pi \times 10^8$ rad/s; figure (b) depicts an *equivalent circuit* in terms of input impedance of the transmission line at $d = l$, namely $Z(l) \equiv \frac{V(l)}{I(l)}$.



(a) Transmission line circuit



(b) Equivalent circui

In answering the following questions assume that the circuit above is in sinusoidal steady-state:

a) (1pt) What is the signal wavelength λ (in m) on the transmission line?

b) (1pt) Since an "open termination" is located on the line at $d = 0$, what is the pertinent "boundary condition" involving the phasor $I(0)$ for all possible line lengths l?

c) (1pt) Does the transmission line support a *standing wave* in the above circuit for all values of non-zero l? Justify your answer.

d) (1pt) What is the smallest non-zero value of l (in m) if phasor $I_R = 2\angle 0$ A? Explain.

e) (1pt) For l determined in part (d), what is phasor $V(l/2)$? Explain.

f) (1pt) For l determined in part (d), is $I(l/2) = 0$ possible? Explain.

g) (1pt) What is the smallest non-zero value of l if $I_R = 0$?

h) (1pt) Given that $I(l/2) = 0$, what is the smallest possible value of l? Explain

i) (2pts) For l determined in part (h), what is $V(l)$? Explain.

3. (10pts) A quarter-wavelength long transmission line section is terminated by an unknown impedance Z_L at one end. The input voltage of the quarter-wave section at the other end is $V_{in} = 100\angle 0^o$ V (in phasor notation). Also the characteristic impedance of the line is $Z_o = 50\,\Omega$.

   Let I_{in} and I_L denote input and load current phasors, respectively, with current directions defined in a compatible way with one another and with V_{in}:

   a) (5pts) Assuming that the load is not open (i.e., $Y_L = \frac{1}{Z_L} \neq 0$) what is I_L?

   b) (2pts) Is $V_{in} = 100\angle 0^o$ V a possibility if the load is an open? Discuss your answer.

   c) (3pts) What is I_{in} if $Z_L = 100\,\Omega$?

4. (10pts) In the transmission-line circuit shown below the voltage at the load is $V_L = j15$ V The operating frequency is 300 MHz. Given that the line is lossless of characteristic impedance of 75 Ohm with wave velocity 3×10^8 m/s, calculate the following:

   a) (3pts) The time-average power dissipated at the load;

   b) (7pts) The time-average power delivered by the generator.

